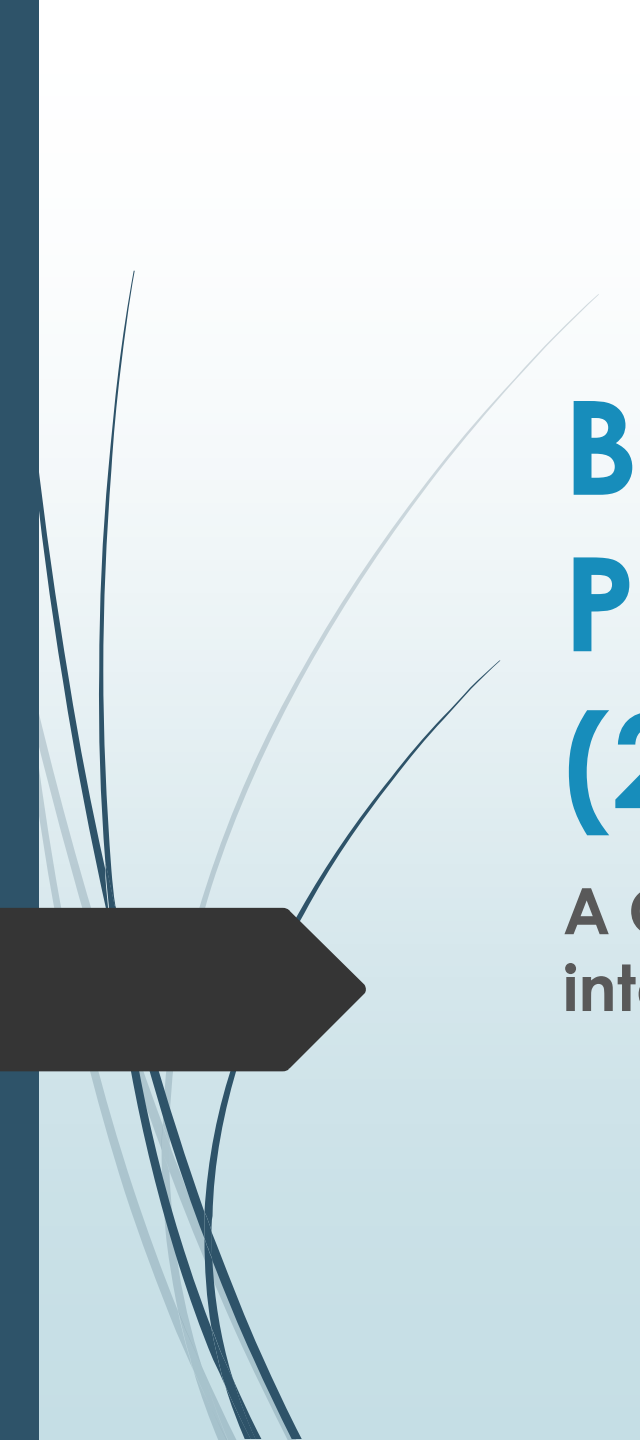




Block Production & Sales Performance Analysis (2021–2025)

**A Comprehensive Data Analytics Report
integrating Excel, Power BI, Stata, and SPSS**

By

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A. EXECUTIVE SUMMARY (INTEGRATED ANALYSIS REPORT)

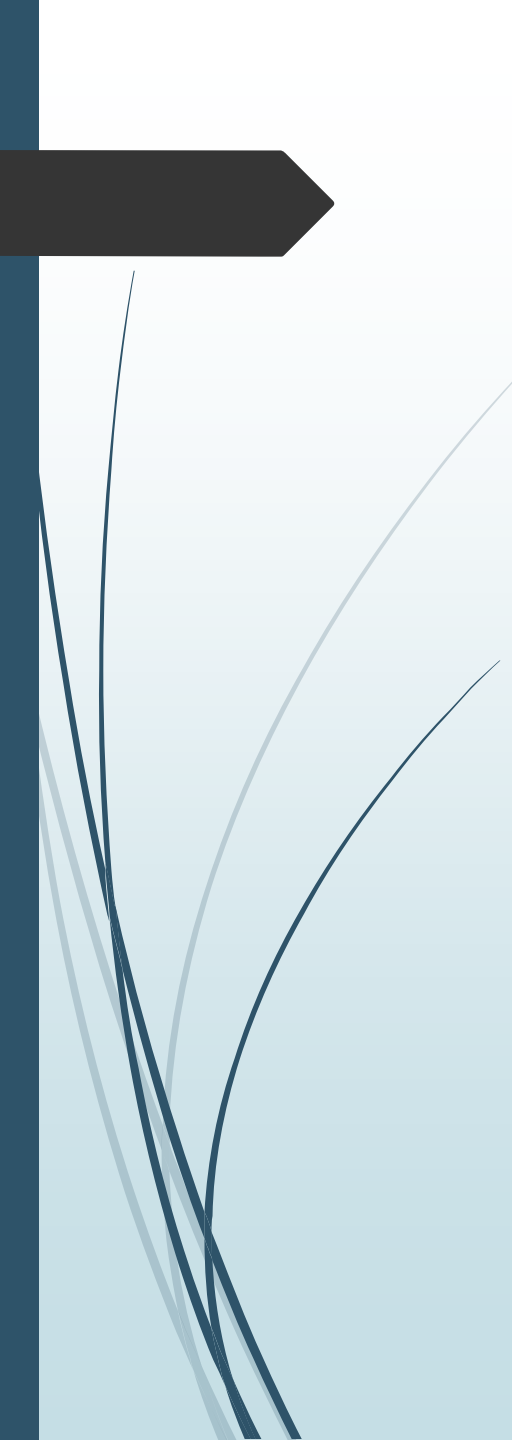
- This report presents a data-driven analysis of production, sales performance, cost structure, and profitability of a block manufacturing business (2021–2025).
- Findings show strong demand and a stable revenue model driven by sales volume and pricing. However, rising production costs and declining operational efficiency are constraining profitability.
- Statistical analysis confirms that while revenue is highly predictable, profitability is influenced by multiple factors particularly cost efficiency rather than production scale alone.



B. BUSINESS PROBLEM & OBJECTIVES



This project aims to analyze production efficiency, sales performance, cost structure, and profitability to identify key drivers of business performance and provide data-driven recommendations.



C. DATA & METHODOLOGY

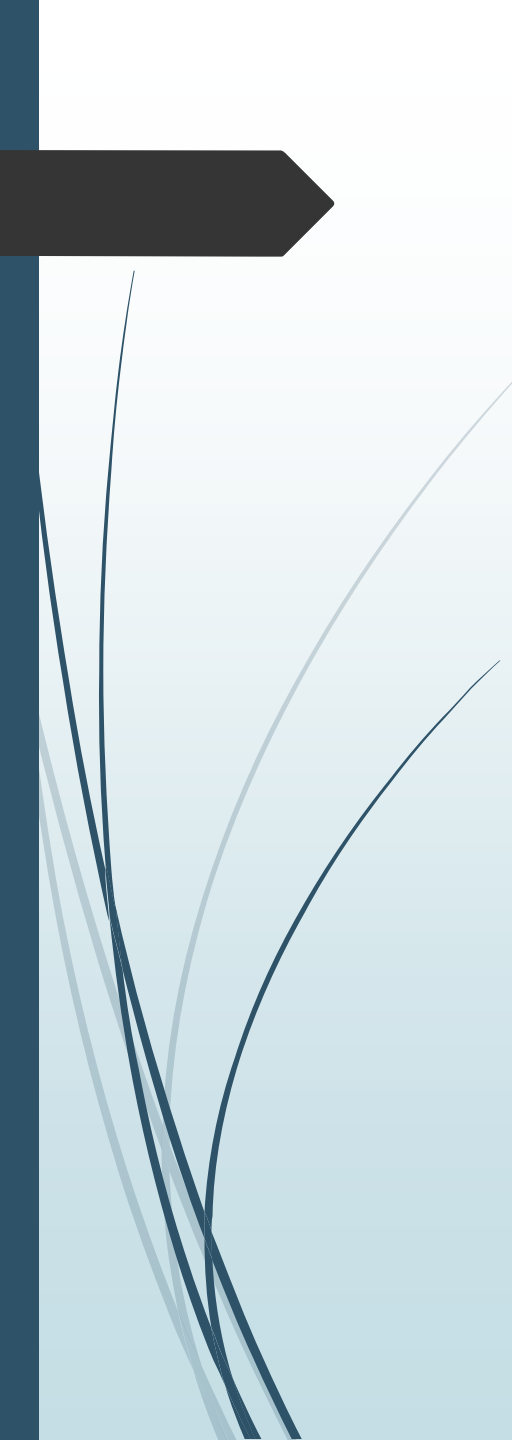
- **Data Source: Internal business records (2021–2025).**
- **Descriptive analysis was conducted using Excel Pivot Tables, while inferential analysis was performed using regression (STATA) and correlation (SPSS). Power BI was used for dashboard visualization.**



D. ANALYSIS & INSIGHTS (CORE SECTION)

◆ 1. PRODUCTION, DEMAND & SALES DYNAMICS

Production trends indicate peak performance in 2022–2023, followed by a decline in 2024 and partial recovery in 2025. Despite these fluctuations, correlation analysis shows a strong and statistically significant relationship between production and sales ($r \approx 0.85$, $p < 0.01$).



Statistical Insight:

> Production has a weak relationship with profit, emphasizing that output expansion without efficiency gains does not improve financial performance.

Interpretation:

Increasing output does not translate into profitability.

Implication:

Efficiency—not scale—is the primary driver of financial performance.



◆ 2. MATERIAL & LABOUR EFFICIENCY

Operational analysis shows:

- Declining labour productivity
- Rising operator cost per bag unit
- Sand inefficiencies (wastage, measurement inconsistencies)



Statistical Reinforcement:

Regression confirms:

- Labour and cost-related variables significantly reduce profit.

Insight:

- > The business is experiencing efficiency decay, where higher input usage does not translate into proportional output gains.



◆ 3. PRICING DYNAMICS

Block prices have increased significantly year-on-year, driven by cost-push inflation. Average block prices grew by up to 51% between 2023 and 2024.

Despite this increase, pricing remains relatively stable within each year.

Key Insight:

> The business has successfully passed rising costs to customers; however, sustained price increases may impact future demand elasticity.



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◆ 4. CUSTOMER STRUCTURE & CREDIT RISK

Descriptive insights show:

- Revenue concentration among few customers
- Mostly low-frequency buyers
- Bad debt $\approx$ 1% of sales

 **Statistical Support:**

Revenue vs Bad Debt → weak & insignificant relationship

 **Insight:**

> Credit risk is present but not a major driver of financial performance; however, customer concentration introduces strategic risk.

◆ 5. PROFITABILITY DRIVERS (CORE FINDING)

The business generated a total profit of ₦3.17 million over the period, with peak performance in 2022.

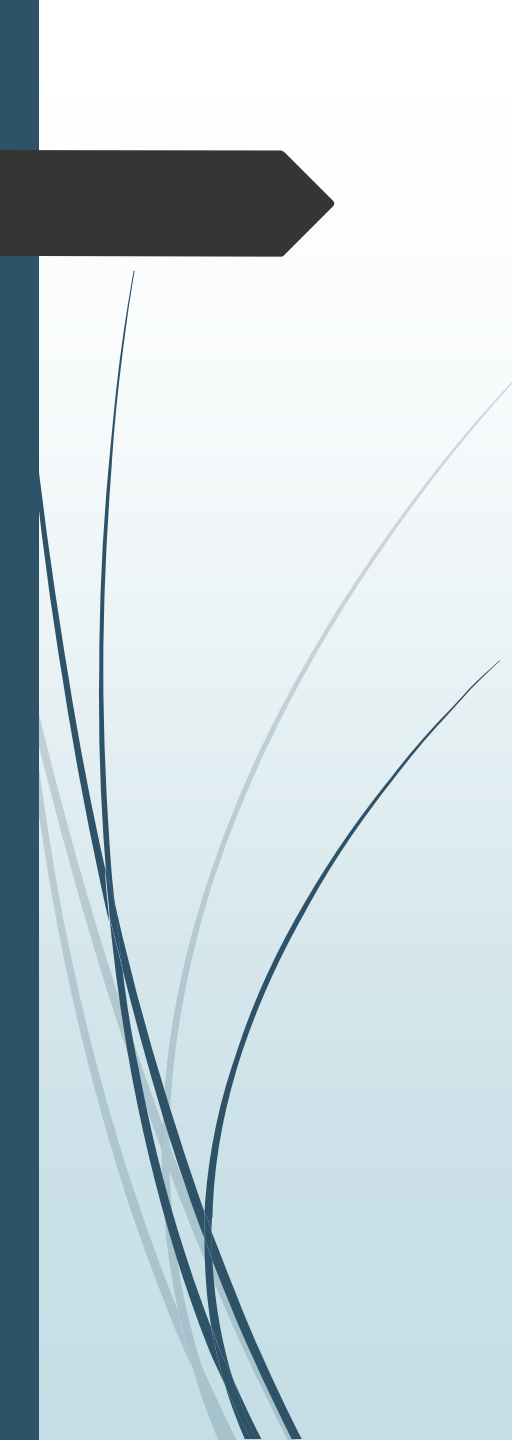
Profit margins remain moderate at approximately 17%, but fluctuate significantly due to cost instability.

Regression model shows:

Total Profit = f (Total Qty Sold, Total Cost, Bad Debt)

Key Results:

- Sales Volume → Positive & Significant
- Total Cost → Negative & Significant
- Bad Debt → Not Significant



Integrated Insight:

> Regression results show that sales volume positively influences profit, while total cost significantly reduces profitability. Bad debt does not have a statistically significant impact.

Interpretation:

- *Selling more increases profit ✓*
- *Rising cost reduces profit ✗*
- *Credit risk is controlled (low impact)*

◆ 6. PRODUCTION VS PROFIT PARADOX (CRITICAL INSIGHT)

Correlation shows:

- Production vs Profit → weak relationship

🔍 Combined Insight:

> Increasing production does not guarantee increased profit.

🎯 Why?

Because:

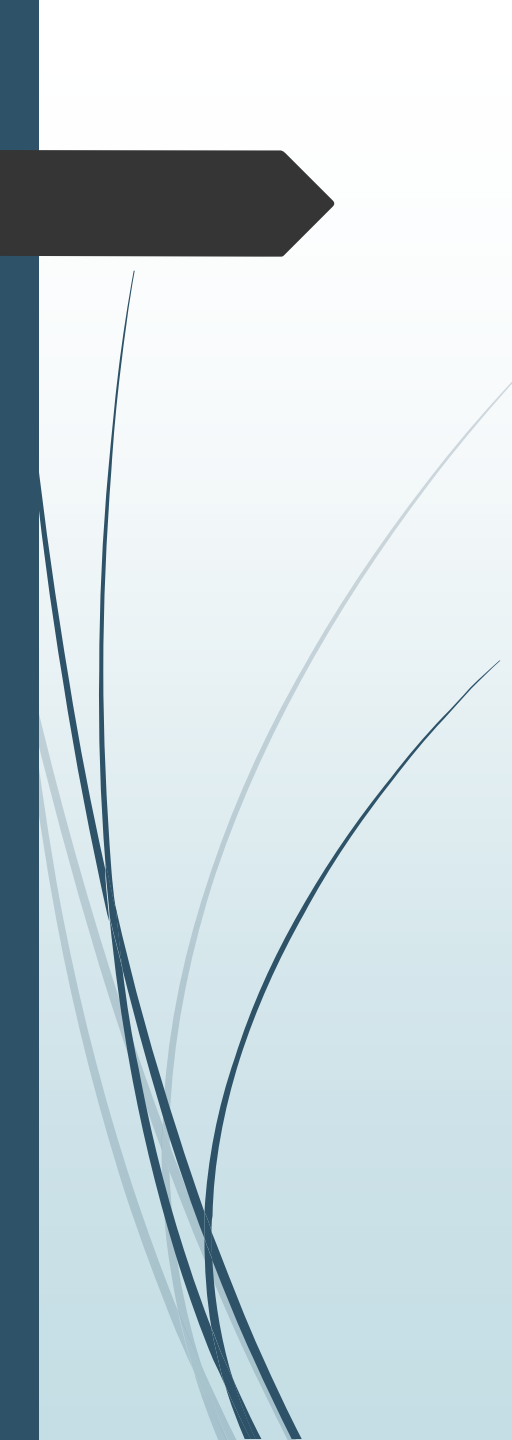
- *Production* ↑ → *Cost* ↑
- *“Profit depends on cost efficiency, not output volume”.*

Correlation Between Key Business Variables

METRIC PAIR	VALUE	STRENGTH
Production vs Sales	0.85	Strong Positive
Production vs Cost	0.83	Strong Positive
Production vs Revenue	0.82	Strong Positive
Production vs Profit	0.09	Very Weak
Cost vs Profit	-0.30	Weak Negative
Revenue vs Bad Debt	0.37	Moderate

Caption Insight:

> Strong relationships exist between productions, sales, cost, and revenue, while a weak relationship exist between production and profit. Also, cost shows a negative relationship with profit, reinforcing the importance of efficiency over volume. And the relationship between revenue and bad debt is weak and statistically insignificant, indicating that credit risk is currently well controlled.



◆ 7. COST STRUCTURE & OPERATIONAL PRESSURE

Descriptive analysis shows a sharp increase in total production cost and cost-per-unit—from ₦148.8 (2021) to over ₦546.7 (2025). Sand and cement dominate cost composition, while labour, delivery, and maintenance costs have grown over time.

Correlation results confirm:

- Production vs Cost → Strong positive ($r \approx 0.84$)
- Cost increases directly with production scale

Regression analysis further establishes:

- Total cost has a significant negative impact on profit ($p < 0.001$)

Integrated Insight:

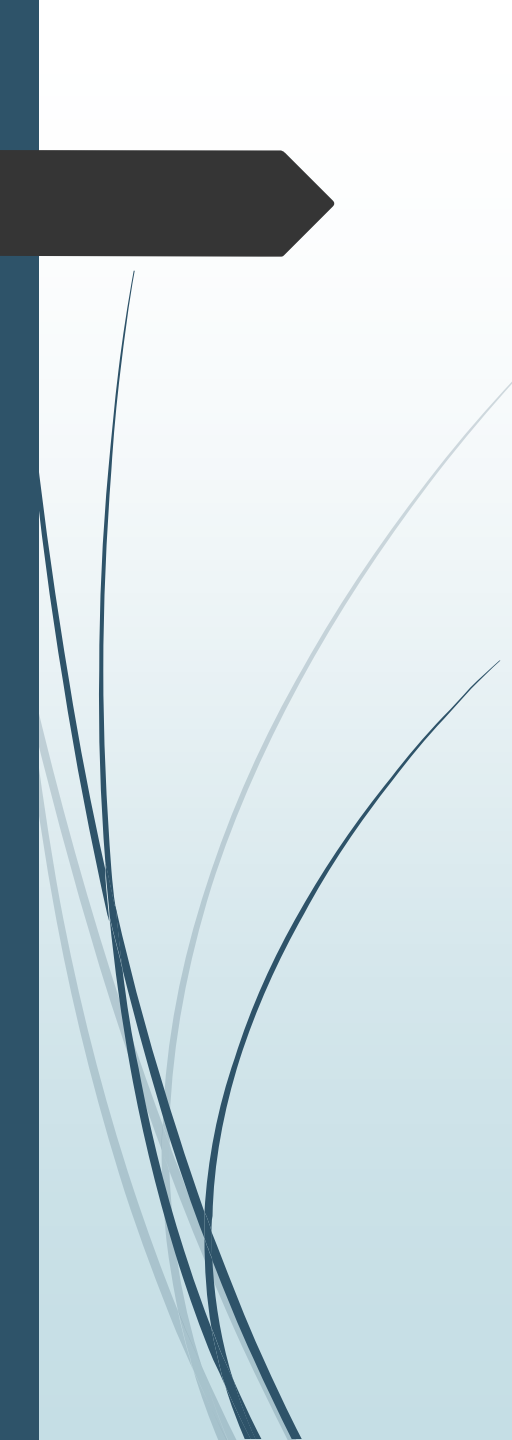
> While scaling production increases revenue, it simultaneously escalates costs—making cost control the most critical determinant of profitability. i.e Profitability declines as production costs increase, confirming cost escalation as the primary constraint on business performance.

Impact of Production Cost on Profitability



Caption Insight:

> Production cost has a statistically significant negative effect on profit; however, it explains only a small proportion of profit variation ($R^2 = 0.087$), indicating that profitability is influenced by multiple interacting factors beyond cost alone.



◆ 8. SALES PERFORMANCE & DEMAND PATTERN

Sales analysis indicates that 6-inch and 9-inch blocks dominate total volume, with demand peaking during key construction periods—particularly October, August, and April.

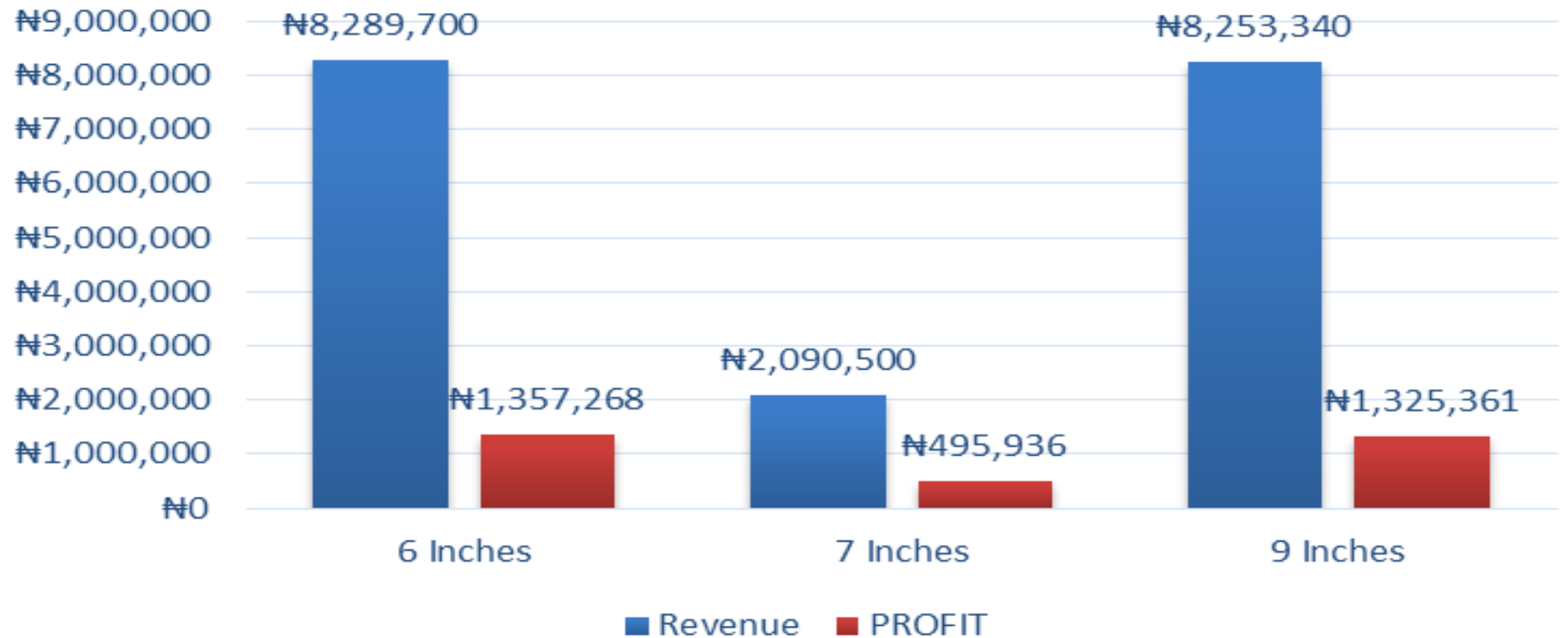
Weekly trends show higher demand during weekdays, aligning with construction activity cycles.

Interestingly, 6-inch blocks deliver the highest profitability per unit, despite not being the highest-priced product.

Key Insight:

> Sales are highly seasonal and predictable, creating an opportunity for demand-driven production planning.

Product Contribution to Sales and Profit



Caption Insight:

> 6-inch blocks generate the highest profitability per unit, also contribute the largest sales volume and revenue.

◆ 9. REVENUE GENERATION MECHANISM

The revenue regression model:

Revenue = f (Qty Sold, Rate)

Shows:

- $R^2 \approx 0.91$ (very strong model)
- Both Qty Sold and Price (Rate) are highly significant ($p < 0.001$)

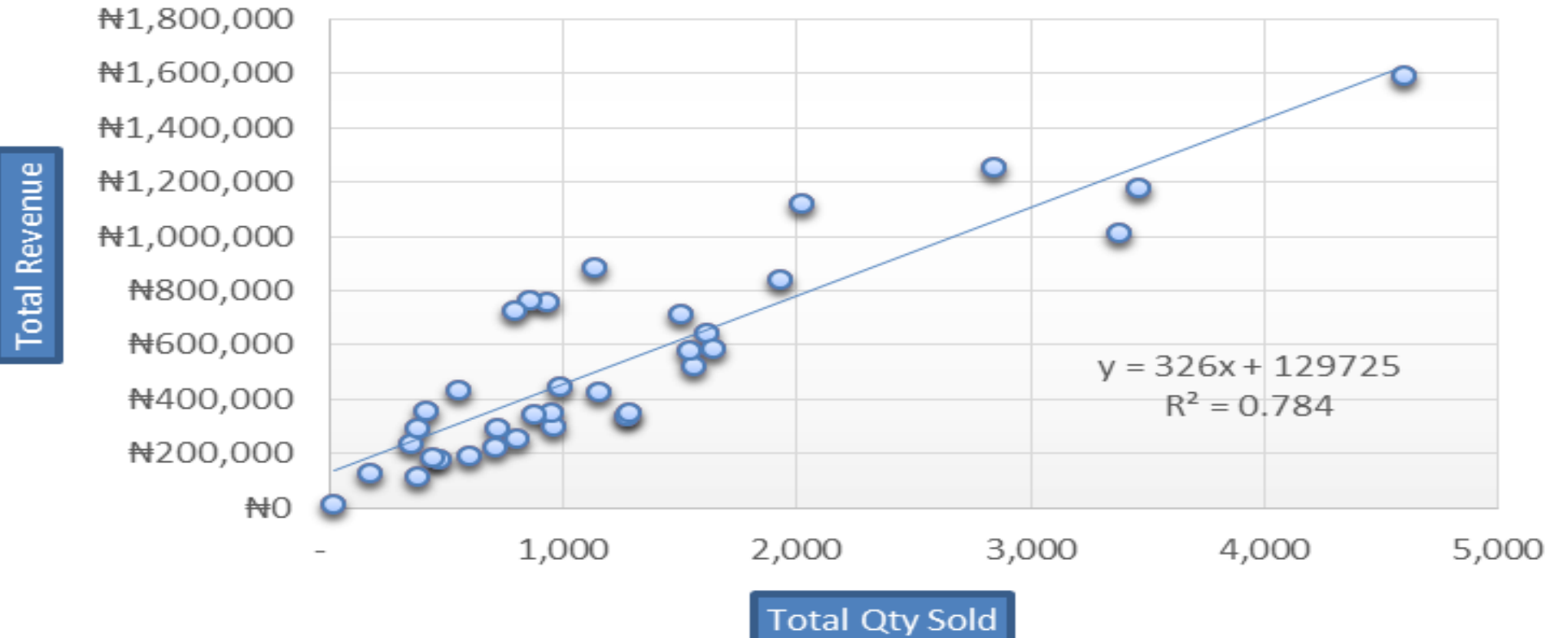
 **Integrated Insight:**

> Revenue is strongly and predictably driven by sales volume and pricing, with regression analysis explaining over 90% of revenue variation. This confirms that increasing sales quantity is the most effective strategy for revenue growth.

 **Business Meaning:**

- *Increasing sales volume = immediate revenue growth*
- *Pricing strategy remains effective despite inflation*

Revenue Growth Driven by Sales Volume



Caption Insight:

> Revenue growth is strongly driven by sales volume, with each additional unit contributing approximately ₦326 to revenue ($R^2 = 0.784$), indicating a high predictive relationship between quantity sold and revenue.

◆ 10. TIME TREND & BUSINESS STABILITY

Time-series regression shows:

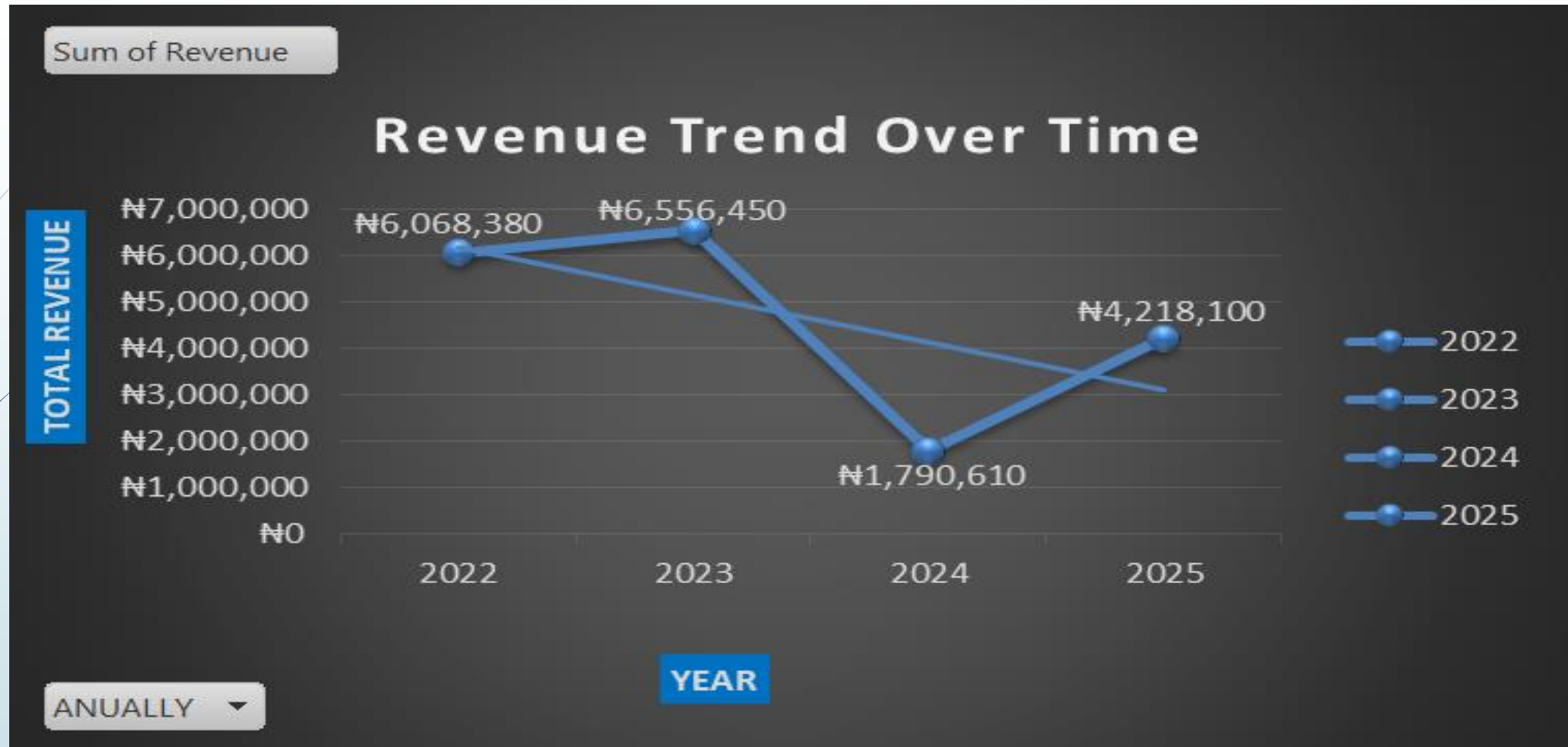
- Date → NOT statistically significant ($p > 0.05$)

 **Insight:**

> Time does not significantly influence revenue, indicating fluctuating rather than steady growth.

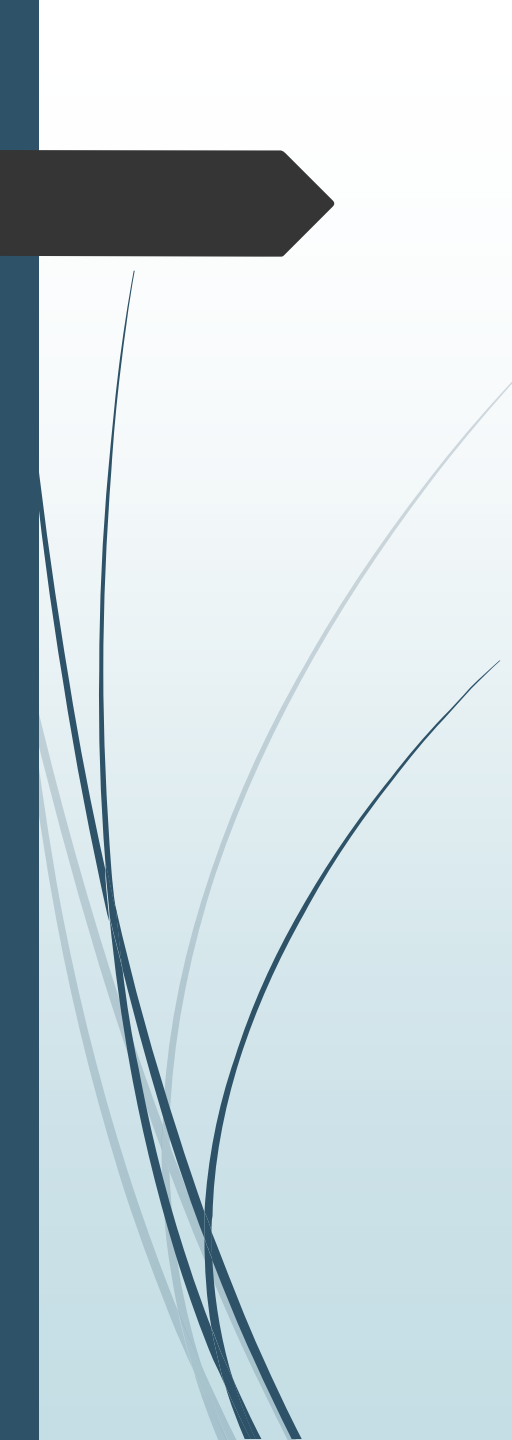
 **Meaning:**

- *Business performance fluctuates*
- *Growth is not stable or predictable*



Caption Insight:

> Revenue fluctuates over time without a consistent upward trend, indicating instability in business performance and the need for improved planning.



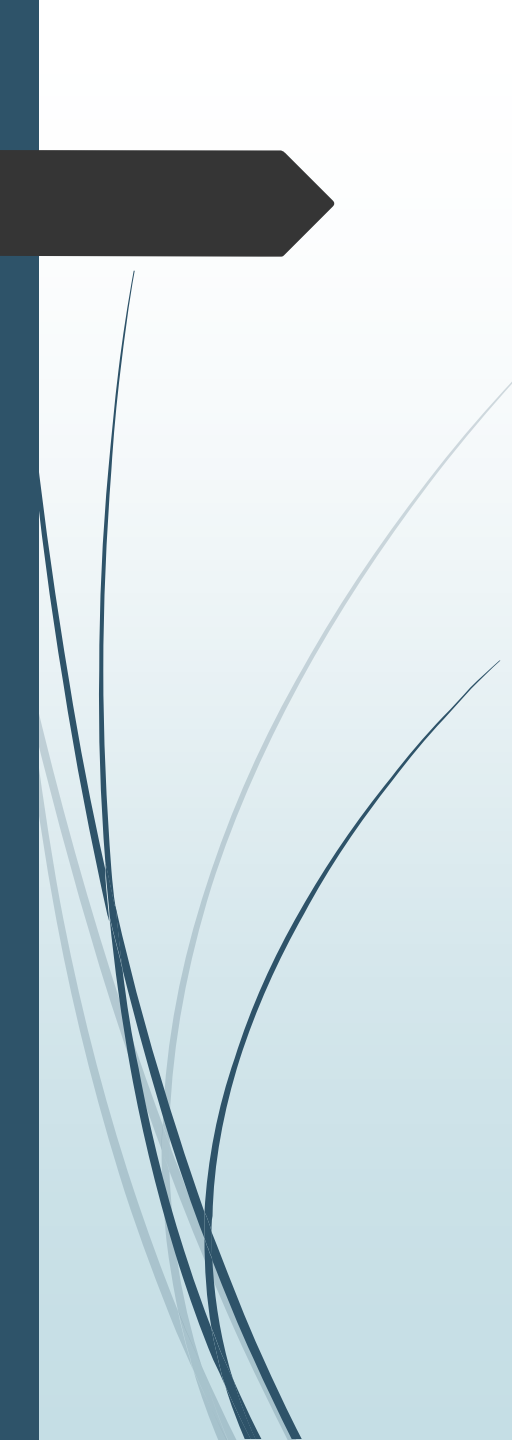
E. OVERALL BUSINESS DIAGNOSIS

✓ Strengths:

1. Strong demand (high production-sales alignment)
2. Stable revenue model (volume + pricing)
3. Low inventory wastage
4. Controlled bad debt

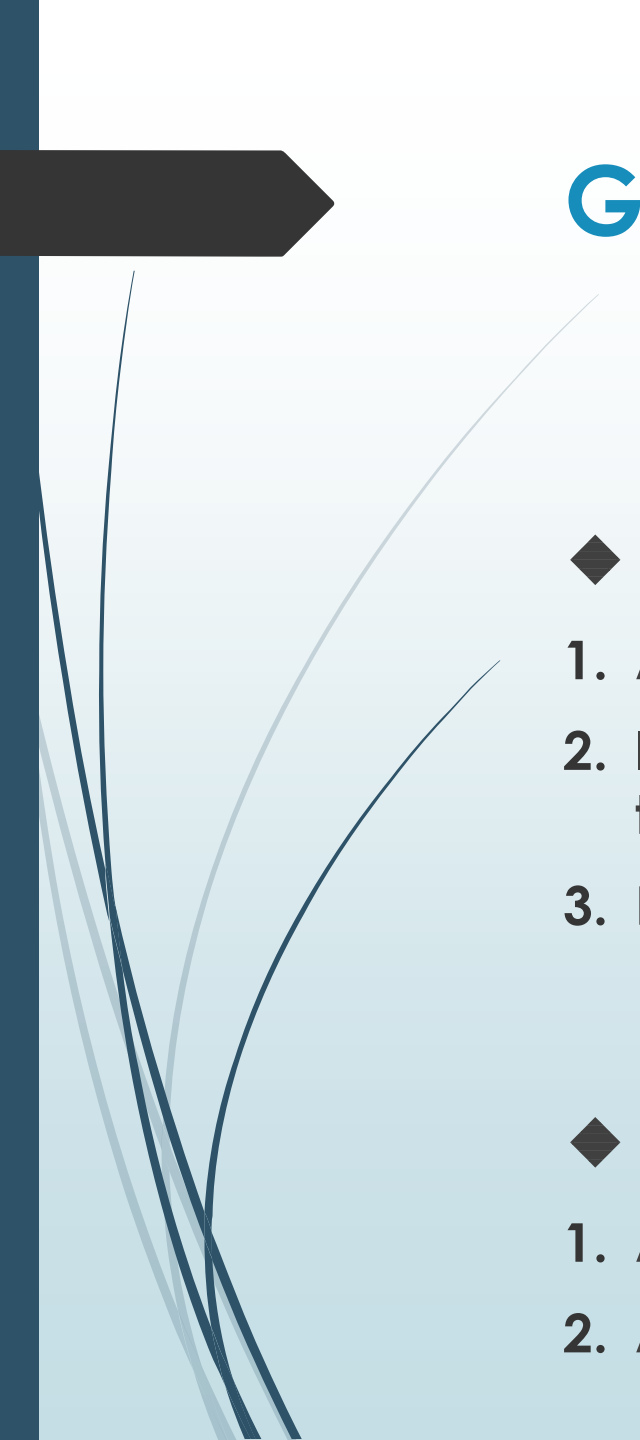
⚠ Weaknesses:

1. Rising cost structure
2. Declining efficiency
3. Unstable growth pattern
4. Customer concentration risk



F. STRATEGIC IMPLICATIONS (DATA DRIVEN)

- 🎯 1. Profit depends on cost efficiency, not production volume
- 🎯 2. Cost escalation is the biggest threat
- 🎯 3. Revenue is driven by sales volume and pricing
- 🎯 4. Business performance is unstable over time
- 🎯 5. Customer concentration introduces risk



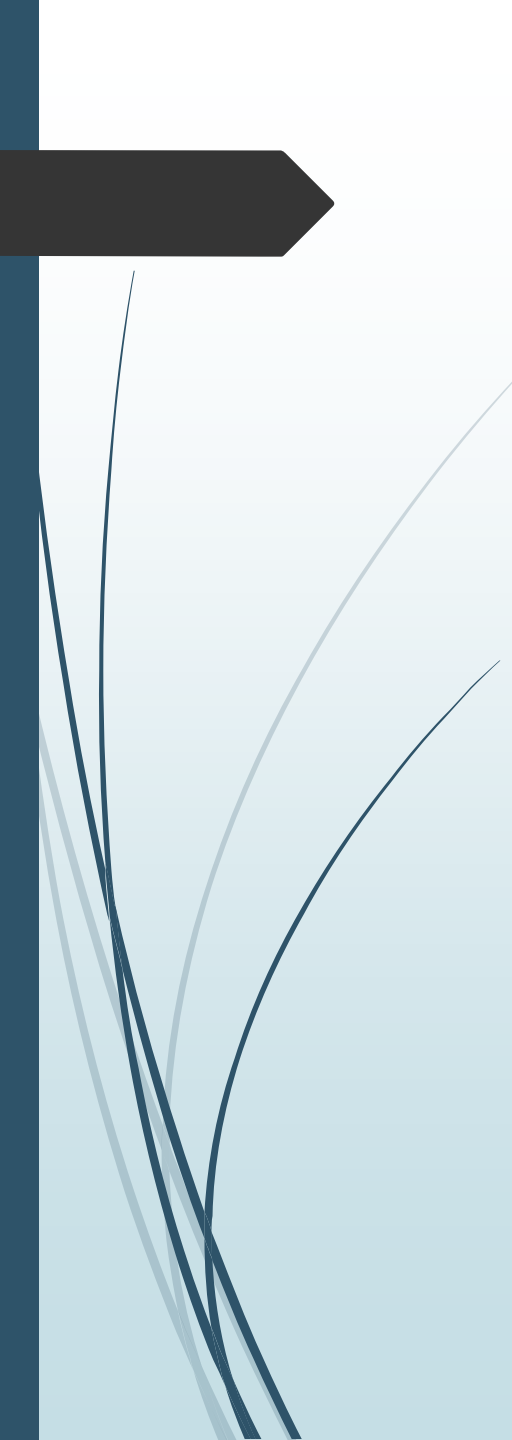
G. STRATEGIC RECOMMENDATIONS (UPGRADED)

◆ Cost Optimization

1. Aggregate cost tracking (Total Cost KPI)
2. Implement material usage benchmarks (cement & sand per block) and track variance to reduce wastage
3. Negotiate supplier pricing

◆ Production Planning

1. Align production with seasonal demand
2. Avoid overproduction cycles



◆ Operational Efficiency

1. Monitor labour productivity
2. Standardize material usage
3. Reduce wastage

◆ Revenue Optimization

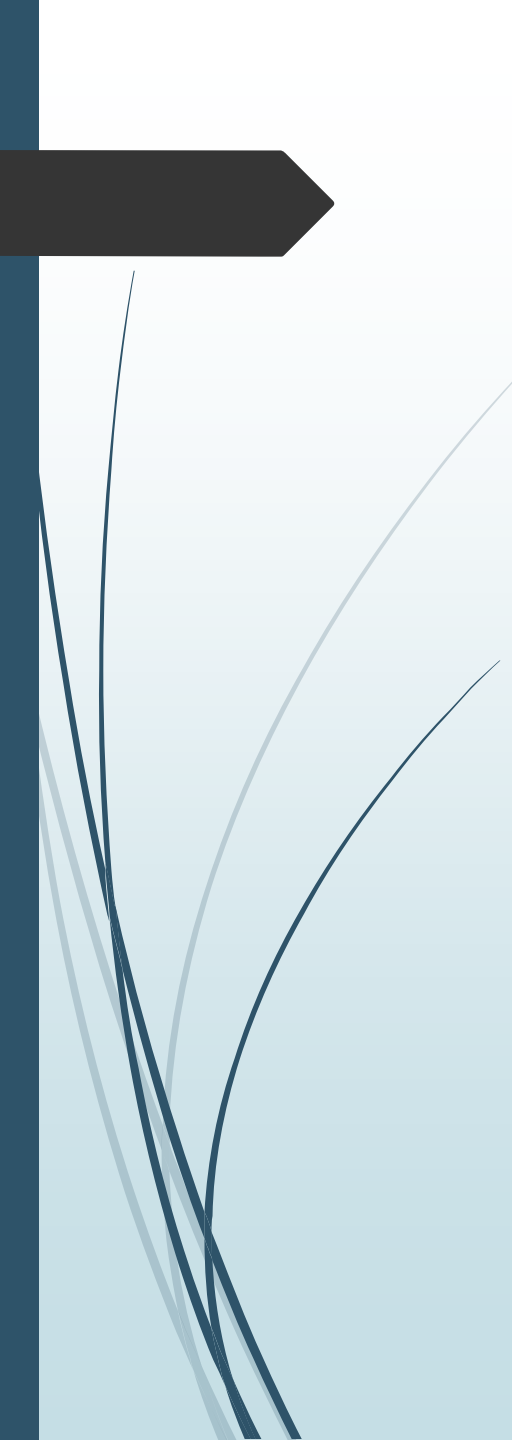
1. Increase high-demand product supply (6-inch blocks)
2. Maintain pricing discipline

◆ Customer Strategy

1. Retain high-value customers
2. Expand customer base

◆ Credit Control

1. Enforce credit limits
2. Track customer-level debt

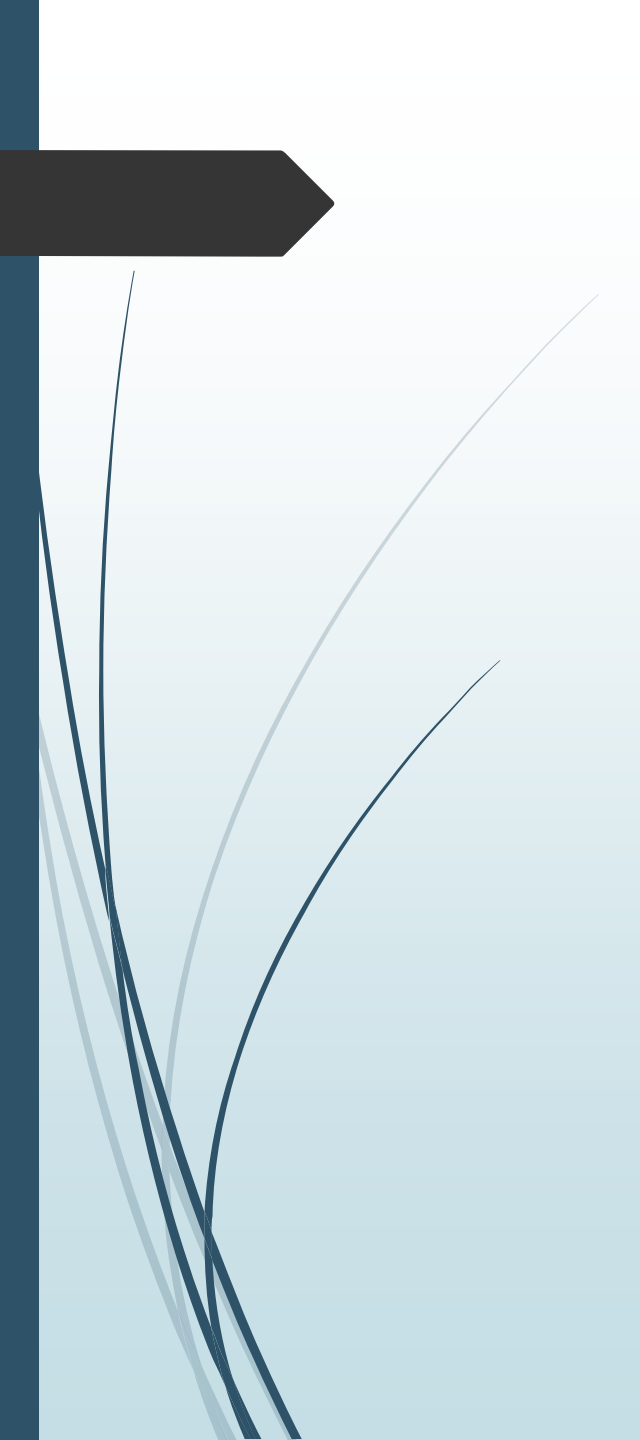


H. FINAL CONCLUSION

The business operates in a high-demand but cost-sensitive environment. While revenue generation is strong and predictable, profitability is increasingly constrained by rising costs and declining efficiency.

The integration of statistical and descriptive analysis confirms that:

> Sustainable growth depends on controlling cost, improving efficiency, and stabilizing operations—not merely increasing production.



THANKS FOR READING!